

Bridge Day 11 STUDENT QUIZ

While Loops, Update Order, and Termination Evidence

Learning sequence: Predict - Trace - Explain - Test - Interpret

Monday, July 20, 2026 • 20 points • target 18 minutes

Name		UIN		Section	
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Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: input conversion, conditions, while, update order, termination

Scaffold level: complete bidirectional while-loop program

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.19, 18.21, 18.22.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Trace a while loop that crosses its target. - 5 points

```
current = 2
target = 7
steps = 0

while current < target:
    current = current + 2
    steps = steps + 1
    print(current, steps)
```

Question: Give the complete pass ledger and explain why the final value can exceed target.

Item 2. Diagnose a non-progressing loop. - 4 points

```
value = 3

while value < 10:
    print(value)
    value + 2
```

Question: Explain why the loop does not terminate, give the minimal repair, and list repaired output.

Complete program - 11 points

- Read start and target as integers. Valid floors are 1 through 10 inclusive.
- Print invalid if either floor is outside the interval and do not run a loop.
- Print already there if the two valid floors are equal.
- Otherwise use a while loop to move exactly one floor per pass toward target and print each newly reached floor.
- After movement, print Steps: followed by the number of loop passes.

Program workspace**Test input, expected result, and why this test matters**